microscopic dust motes in a fluid. The botanist Robert Brown had noticed this phenomenon nearly eighty years earlier (and Jan Ingenhousz, the discoverer of photosynthesis, had seen it forty years before that) but nobody had been able to satisfactorily explain it. Einstein did so masterfully—and he did it by rejecting Mach's approach to physics. Instead, Einstein adopted the approach of Mach's nemesis, Ludwig Boltzmann, who claimed that the world was made of a phenomenal number of tiny atoms. Mach had loudly and repeatedly proclaimed that he did not believe in atoms, as they were too small to be observable in principle. But Boltzmann had managed to show that the statistical behavior of massive numbers of atoms led directly to the laws of thermodynamics that Mach had been so eager to simply assume. (There was also evidence for atoms from chemistry, which had by then accepted the existence of atoms for over half a century.) Mach was unconvinced by Boltzmann's arguments. But Einstein found them compelling and elegant, and happily subscribed to the existence of atoms in order to solve the problem at hand. Using Boltzmann's statistical methods, Einstein showed that Brownian motion was caused by the dust motes bouncing off of the atoms in the fluid. In one stroke, Einstein not only explained a century-old puzzle, but conclusively demonstrated that Boltzmann's statistical, atom-based approach to physics was both sound and useful.

As bad as Einstein's Brownian motion paper was from a Machian perspective, his other paper was even worse. In it, Einstein again proposed a solution for an old puzzle, the photoelectric effect, in which shining light on a metal plate could cause a current to jump through the air to a nearby wire. The puzzling thing about the photoelectric effect was that the color of the light involved seemed to matter: if the light was too far toward the red end of the spectrum, then no matter how bright the light was, no current was seen. Einstein accounted for this strange behavior by proposing light was composed of a totally new particle, the photon. This was an audacious hypothesis that not only flew in the face of Machian philosophy but also seemingly contradicted a century of experimental evidence that light was a wave, not a particle. Einstein certainly knew that light was an electromagnetic wave—the idea was crucial inspiration for his theory of relativity—but was nonetheless proposing that light was somehow also a particle, or had some kind of particle-like nature. In defense of this strange